SOFTWARE TEST PLAN
Collaborative Study Material & Quiz Platform

Field	Details
Project	Collaborative Study Material & Quiz Platform
Team No.	16
Version	1.0
Date	16-09-2026
Status	Draft
Reference SRS	Version 1.0, 09-09-2026

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	16-09-2026	Team 16	Initial project-specific test plan based on SRS and supplied STP template	Pending

Approvals

Role	Name	Signature / Date
Course Coordinator	_____	_____
Project Evaluator	_____	_____

1. Introduction

Purpose

This Software Test Plan (STP) defines the objectives, scope, strategy, resources, schedule, responsibilities, entry and exit criteria, security validation, and reporting approach for testing the Collaborative Study Material & Quiz Platform. It adapts the supplied ATM Software Test Plan structure to this project's SRS.

Scope

Testing covers authentication and role-based access, study-material management, collaboration and version history, comments and ratings, quiz creation and attempts, question types, timing and randomization, automatic evaluation, instant results, quiz history, leaderboards, administration, security controls, performance, reliability, usability, data integrity, maintainability, and workload support.

The SRS defines the product as a standalone web application accessed through a modern desktop/laptop browser, with backend services, a database, and file storage. The technology stack is intentionally open at the SRS stage, so this plan avoids assuming a particular framework or vendor.

References

- Collaborative Study Material & Quiz Platform SRS, Team 16, Version 1.0, 09-09-2026.
- Software Architecture and Design Specification, Team 16, Version 1.0.
- Supplied Software Test Plan (STP) template for the Software Engineering mini project.

Definitions

Term	Definition
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role Based Access Control
UAT	User Acceptance Testing
TC	Test Case

2. Test Items

- Authentication and RBAC module
- Study Material Management module
- Collaboration and Versioning module
- Quiz Management and Evaluation module
- Administration and Audit module
- Web UI and supported browser workflows
- Database and persistent records
- File storage and uploaded study documents
- • Security controls and protected interfaces

3. Features to be Tested

The plan covers all 22 functional requirements, 6 non-functional requirements, and 6 security requirements in the SRS. The SRS already associates each requirement with a test-case reference.

Area	Requirement IDs	Planned Test Cases
Authentication & RBAC	CSQP-F-001–002	TC-AUTH-01–02
Study Material	CSQP-F-003–010	TC-MAT-01–08
Collaboration & Versioning	CSQP-F-011–012	TC-MAT-09, TC-COL-01
Quiz	CSQP-F-013–019	TC-QUIZ-01–07
Administration & Audit	CSQP-F-020–022	TC-ADMIN-01–02, TC-SEC-01
NFRs	CSQP-NF-001–006	TC-PERF-01, TC-REL-01, TC-UX-01, TC-DATA-01, TC-MAINT-01, TC-SCALE-01
Security	CSQP-SR-001–006	TC-SEC-02–07

4. Features Not to be Tested

- Internal implementation details of third-party infrastructure outside the team's control.
- Specialized hardware interfaces; the SRS requires no specialized hardware.
- Production-scale behaviour beyond the agreed academic workload.
- Technology-specific automated scripts before the final stack/API contract is selected.

5. Test Approach / Strategy

5.1 Levels

Level	Purpose
Unit Testing	Verify individual application logic components.
Integration Testing	Verify Web UI, backend services, database, and file-storage interactions.
System Testing	Verify complete end-to-end workflows against the SRS.
Acceptance / UAT	Verify SRS acceptance criteria and core user workflows.

5.2 Types

- Functional testing
- Regression testing
- Performance/load testing
- Reliability/availability testing
- Usability testing
- Data-integrity testing
- Maintainability/design review
- Security testing

5.3 Entry Criteria

- Stable test build delivered.
- Test environment ready.
- Representative test data prepared.
- SRS and design documentation available.
- Basic smoke testing does not reveal a blocker to planned execution.

5.4 Exit Criteria

- 100% of planned test cases executed for the release under test.
- All high-priority functional requirements verified or formally excepted.
- No unresolved critical security or data-integrity defect.
- CSQP-NF-001 performance threshold met under agreed load.
- Acceptance criteria satisfied or formally dispositioned.
- RTM coverage complete.

5.5 Security Validation

Requirement	Validation
CSQP-SR-001	Inspect password storage and verify no plaintext passwords.
CSQP-SR-002	Inspect sensitive traffic and verify HTTPS/TLS.
CSQP-SR-003	Attempt protected operations with unauthorized roles/direct requests.
CSQP-SR-004	Submit malformed/unsafe inputs and invalid file metadata.
CSQP-SR-005	Test logout and configured session timeout.
CSQP-SR-006	Attempt client/request manipulation of quiz scores.

6. Test Environment

Category	Planned Environment
Hardware	Standard desktop/laptop; no specialized hardware.
Client	Modern desktop/laptop web browser supported by final implementation.
Application	Collaborative Study Material & Quiz Platform test build.
Backend	Project application services selected during implementation.
Database	Project persistent database selected during implementation.
File Storage	Project file-storage solution.
Network	Internet/institutional network using HTTPS/TLS.
Tools	Browser developer tools; API client if API is exposed; JMeter or equivalent for performance if required; defect tracking tool selected by team.
Test Data	Dummy Student/User, Content Contributor/Quiz Creator, Administrator accounts; sample notes/PDFs, quizzes, answers, comments, ratings and attempts.

7. Test Schedule

Milestone	Planned Date / Status
Test plan and test case design	16-09-2026 — Complete draft
Environment setup	After implementation build is available
Test execution start	To be scheduled
Functional/system execution	To be scheduled
Security and NFR verification	To be scheduled
UAT	To be scheduled
Test closure / summary	To be scheduled

8. Test Deliverables

- Software Test Plan
- Test Cases and Requirements Traceability Matrix
- Test Data
- Test Scripts / automation artifacts, if applicable
- Test Execution Logs
- Defect Reports
- Test Summary Report

Execution logs, defect reports, and final pass/fail results are not fabricated here because the SRS records the implementation status of requirements as N (Not yet implemented). They become execution-phase deliverables.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	To be assigned	Maintain plan, coordinate execution, review coverage, compile summary.
Test Engineer	To be assigned	Design/execute cases, record evidence, log defects.
Developer	Team 16	Support builds, fixes, analysis and triage.
Evaluator / Product Owner	Course Coordinator / Project Evaluator	Review acceptance evidence and sign off.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build	Prepare tests early and request smoke builds.
Environment downtime	Maintain reproducible local test setup.
Technology stack not finalized	Keep cases technology neutral.
Concurrent editing ambiguity	Use SRS controlled update/versioning criteria and define conflict behaviour during implementation.
Insufficient test data	Prepare reusable dummy accounts and content.
Late security defects	Run security validation alongside functional testing.
Non-reproducible performance results	Record hardware, browser, network, concurrent-user target, and dataset size.

11. Assumptions & Dependencies

- Users have valid credentials and a supported browser.
- Persistent storage is available.
- Administrators are available for moderation/account scenarios.
- Deployment provides HTTPS and adequate storage.
- Network connectivity is available.
- Configured file type/size restrictions are defined before validation testing.
- Final technology stack exposes testable workflows/interfaces.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable and prevents meaningful execution.
- Build instability blocks a substantial portion of planned cases.
- A blocking defect prevents a major prerequisite workflow.
- Required test data/dependency is unavailable.

Resume testing if:

- Blocking defects are resolved or formally waived.
- Environment is stabilized.
- Required data and dependencies are available.
- A stable build passes smoke testing.

13. Test Case Management & Traceability

The RTM maps every SRS requirement to one or more planned test cases. The SRS provides references for all 22 functional requirements, 6 NFRs, and 6 security requirements. A separate workbook contains the expanded planned cases and coverage matrix.

Coverage	Count
Functional requirements	22
Non-functional requirements	6
Security requirements	6
Total requirements	34
Planned test cases	34

14. Test Metrics & Reporting

Metric	Definition
% test cases executed	Executed / planned × 100
% passed	Passed / executed × 100
% failed	Failed / executed × 100
Defect density	Defects relative to the selected size measure.
Defect aging	Time defects remain open.
Requirement coverage	Requirements with executed verification / total requirements × 100.
Critical defects open	Unresolved critical defects at a reporting point.

Reports: daily execution status during active testing and a final Test Summary Report at closure.

15. Approvals

Role	Name	Signature / Date
QA Lead	_____	_____
Developer / Dev Lead	_____	_____
Course Coordinator / Evaluator	_____	_____

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